DUY KHUONG

kd23132@seas.upenn.edu | linkedin.com/in/duy-khuong-541320212/ | 215-868-8693 | Philadelphia, PA 19104

EDUCATION

University of Pennsylvania ***GPA: 3.33 / 4.00***
M.S., Mechanical Engineering and Applied Mechanics *Aug 2025 – May 2027*

- Concentration: Mechatronic and Robotic Systems

VinUniversity ***GPA: 3.55 / 4.00***
B.S., Mechanical Engineering *Sep 2022 – May 2026*

- Dean's List, Fall 2023 and Fall 2024; 90% Merit Scholarship for Undergraduate Study

SKILLS

- **Tools**: Python, R, MATLAB, Git, Bash, Autodesk Fusion, SolidWorks, Ansys Workbench, CNC Machining
- **Frameworks & Libraries**: Pytorch, TensorFlow, Numpy, Pandas, FedML, Librosa, MNE, OpenCV

RESEARCH EXPERIENCE

Security and Artificial Intelligence Lab | VinUniversity *Aug 2024 – May 2025*
Undergraduate Research Assistant

- Developed a federated method to quantify each modality's contribution and rebalance training for multimodal classification, outperforming popular methods.
- Evaluated a private multimodal dataset and delivered comprehensive quality/labeling feedback.
- First author of "*FedKoE: Enhancing Federated Multimodal Learning through Knowledge of Experts,*" published in the *FL-AsiaCCS 2025* workshop.

PROJECTS

Project Arduino Based: Self Balancing Robot | VinUniversity *Aug 2024 – Jan 2025*

- Built a two-wheeled self-balancing robot for the Mechatronics (MECE3060) course using IMU sensor integration and control design; enabled autonomous balance and position navigation.
- Led Arduino hardware assembly, designed the 3D-printed frame, and tuned the self-balancing control parameters; only team to finish on time, earned top performance a perfect score (100/100)

Project Quadcopter Design | VinUniversity *Feb 2025 – Jun 2025*

- Built a quadcopter for the MECE3080 (Intelligent Physical Systems) course using aerodynamic design and control theory to achieve stable hover and waypoint navigation.
- Designed and fabricated a carbon-fiber frame through running ANSYS structural simulations, iterating 3D-printed prototypes and optimizing geometry and CNC-machined carbon-fiber parts to achieve top stability and highest score (19/20) among 5 groups.

Project Barrier Gate | VinUniversity *Feb 2025 – Jun 2025*

- Designed and fabricated a motor-driven swing barrier gate using quick-return mechanism for the MECE3040 (Mechanical Synthesis) course. Performed kinematic analysis with MATLAB, CAD modeling, and manufactured components via laser cutting and CNC machining to achieve optimized opening/closing time ratio.

Project Digital Stethoscope | VinUniversity *Aug 2025 – Jan 2026*

- Designed and assembled a 4-piece 3D-printed enclosure and frame for a wireless digital stethoscope, providing mechanical support and safe packaging for the embedded system.
- Ensured reliable Bluetooth communication between the ESP32 and mobile application for real-time PCG streaming and trained a lightweight CNN-based classifier (normal vs. abnormal) using the PhysioNet 2016 Heart Sound Challenge dataset.

Project Robot Battle Arena | University of Pennsylvania *Oct 2025 – Dec 2025*

- Designed and assembled a robot mechanical frame and sensor wiring (Vive tracking) for the MEAM5100 course, integrating a mecanum-drive mobile base and ensuring reliable hardware bring-up for teleoperation.
- Built control stack with a Bluetooth Xbox controller teleoperation and autonomous behaviors for the game's 30 secs autonomous phase, including wall-following navigation and an attack strategy to damage the opponent nexus.